

	Insufficient	Below Expectation	Competent	Effective
Calculating Discount Rates (10%)	[0-2 marks] Discount rates are absent, a single arbitrary rate is used for all cash flows without justification, or the applied method is fundamentally incorrect or inappropriate.	[3-5 marks] A basic approach to determining discount rates is present. The rates used may not adequately account for varying maturities of cash flows, or the source/justification for the rates is simplistic.	[6-8 marks] An attempt is made to derive discount factors that vary with the timing of cash flows, using observable market information, and the methodology is generally sound.	[9-10 marks] A robust method is implemented to derive a consistent set of discount factors for all relevant future periods. The approach reflects careful consideration of appropriate data sources.
Asset Valuations (20%)	[0-5 marks] Valuations for some or all specified derivative trades are missing or contain fundamental errors. Methods used are largely inappropriate for the instruments being valued.	[6-10 marks] Valuations are attempted for most specified derivative trades, but the methodologies applied may be overly simplistic. Accuracy of the derivative prices is limited.	[11-15 marks] Valuations are correctly produced for all specified derivative trades using methods generally appropriate for each instrument's characteristics.	[16-20 marks] Highly appropriate, well-justified, and accurately implemented valuation methodologies are applied throughout the framework of the software.
Hedging Analysis (15%)	[0-3 marks] Portfolio hedging parameters are not calculated, or the calculations are fundamentally flawed. No meaningful risk assessment is provided.	[4-7 marks] At least one basic risk parameter is calculated for the portfolio, but the calculation may contain errors. The scope of hedging analysis is very limited.	[8-11 marks] A robust set of key relevant risk parameters is computed for the portfolio. The analysis demonstrates insight into their practical application for managing portfolio risk.	[12-15 marks] A comprehensive suite of risk parameters is accurately determined. The hedging information provided extends to actionable instructions that allow for the effective management of identified risks.
Object-Oriented Architecture (20%)	[0-5 marks] Object-Oriented principles are largely absent or applied incorrectly.	[6-10 marks] A basic attempt at OOP is made with classes for some instruments, but the design shows limited understanding or flawed application of core OOP concepts.	[11-15 marks] OOP principles are clearly utilised with distinct classes for the specified derivatives and their components. The codebase is moderately organised and reusable.	[16-20 marks] A sophisticated and well-reasoned OO architecture is implemented. The design results in a flexible, extensible, and maintainable system with minimal code duplication and clear separation of concerns.
Research & Data Sourcing (10%)	[0-2 marks] Necessary market data and model parameters are largely missing or hardcoded without clear basis.	[3-5 marks] Some attempt is made to source external data for key inputs, but choices may be simplistic, poorly justified, or data may be incomplete or stale.	[6-8 marks] Most required market data are sourced from credible external locations. Some choices made demonstrate external research.	[9-10 marks] All necessary market data and model parameters are appropriately sourced from reliable sources, and choices and assumptions made demonstrate comprehensive external research.
Solution Design & Critical Reasoning (15%)	[0-3 marks] There is little to no explanation or justification for the chosen methodologies, models, or parameters. Design choices appear arbitrary.	[4-7 marks] Basic explanations for some design choices are provided, but reasoning is often superficial, unclear, incomplete, or contains conceptual flaws.	[8-11 marks] Key design decisions and significant assumptions are documented and generally well-justified. The solution demonstrates a logical flow and a proficient understanding of core financial concepts.	[12-15 marks] Demonstrates deep critical thinking through insightful and clear justifications for all significant design choices, model selections, and parameter estimations.
Overall Code Quality (10%)	[0-2 marks] Code is extremely difficult to follow, lacks comments, is highly redundant, or contains significant logical errors.	[3-5 marks] Code is somewhat difficult to follow, has minimal comments, contains some redundancy, or minor logical issues.	[6-8 marks] Code is generally easy to follow, reasonably commented, mostly efficient, and functions correctly.	[9-10 marks] Code is clear, well-commented, efficient, follows good programming practices, and is easy to understand and maintain.